MILITARY SPECIFICATION

FUZE, PROXIMITY, DISPENSER FMU-140/B

This specification is approved for use within the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification establishes the requirements for manufacture and acceptance of the FMU-140/B Dispenser Proximity Fuze, referred to herein as the "fuze".

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Standards. The following standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addresed standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1325

AMSC N/A

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels.

MILITARY

MIL-STD-105	Sampling and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-331	Fuze and Fuze Components, Environmental and Performance Tests for.
MIL-STD-454	Standard General Requirements for Electronic Equipment.
MIL-STD-810	Environmental Test Methods and Engineering Guidelines.
MIL-STD-1167	Ammunition Data Card.
MIL-STD-1322	Unit loads of Ammunition and Explosives for Domestic and Overseas Shipment.
MIL-STD-1695	Environments, Working, Minimum Standards for.
MIL-STD-45662	Calibration Systems Requirements.

2.1.2 Drawings and publications. The following drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

Naval Air Systems Command
(Code Ident 30003)

1472AS200	Fuze, Proximity, Dispenser FMU-140/B.

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1472AS400	Packaging and Marking for Two FMU-140/B Fuze Assemblies.

PUBLICATIONS

Naval Air Systems Command
(Code Ident 30003)

WS-6536	Procedures and Requirements for Preparation and Soldering of Electrical Connections.

Code of Federal Regulations (CFR)

49 CFR 171-178	Transportation.

(Copies of standards, drawings, and publications, required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1 Item description. The fuze shall be a self-powered, active single-mission, radio frequency (RF), radar altimeter device capable of safing, arming, and operating in a high electronic countermeasures (ECM) environment and capable of providing a proximity or time function mode for dispenser weapons.

3.2 Characteristics.

3.2.1 Performance.

3.2.1.1 Mechanical.

3.2.1.1.1 Fuze initiation. The fuze shall initiate (see 6.3.1) when an axial pull force of 125 ±10 pounds is applied to the fuze primary arming wire that is connected to the fuze battery firing device.

3.2.1.1.2 Time function mode select. The time function mode shall be selected when an axial pull force of 125 ±10 pounds is applied to the option arming wire in conjunction with fuze initiation (see 3.2.1.1.1).

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3.2.1.1.3 Air velocity sensor. The fuze shall have an air velocity sensor which prevents arming at velocities less than 160 knots calibrated airspeed (KCAS) and allows arming at velocities greater than 225 KCAS.

3.2.1.1.4 Seal integrity. When the fuze is pressurized and immersed in alcohol, there shall be no evidence of leakage as indicated by a steady stream of gas bubbles for an observation period of 30 seconds minimum. A steady stream of gas bubbles is defined as the formation of a bubble at the fuze prior to the last bubble reaching the surface of the alcohol.

3.2.1.1.5 Explosive output. The fuze shall provide an explosive output which will create a dent of not less than 0.032 inch depth in a dent block (see 6.3.7).

3.2.1.1.6 Operating life. The fuze operating life shall be not less than 60 seconds after fuze initiation (see 3.2.1.1.1).

3.2.1.2 Electrical.

3.2.1.2.1 Proximity function mode. The proximity function mode shall be selected by pulling only the fuze arming wire.

3.2.1.2.1.1 Arm time. The arm times from fuze initiation to the arm pulse shall be in accordance with Table I.

3.2.1.2.1.2 Height of function (HOF).

3.2.1.2.1.2.1 Primary HOF. The fuze shall have 10 manually-selectable HOF settings as specified in Table II.

3.2.1.2.1.2.2 Alternate HOF. The fuze shall provide the output of 3.2.1.1.5 between the primary HOF and 250 feet.

3.2.1.2.1.3 Loop loss sensitivity. When operated in an environment that simulates a closing velocity of 1,200 +100, -0 feet per second (fps), the loop loss (see 6.3.2) that causes the output of 3.2.1.1.5 to occur shall not be less than 122 decibels (dB) except after exposure to the environmental conditions of 3.2.2, the minimum loop loss shall be 120 dB.

3.2.1.2.1.4 Operating frequency (f_1, f_2). The operating frequencies are defined in Appendix I and are provided under separate cover.

3.2.1.2.1.5 ECM immunity.

3.2.1.2.1.5.1 ECM high level immunity. The application of a swept continuous wave (CW) jamming signal from f_1 - 10 megahertz (MHz) to f_1 + 10 MHz at a rate not to exceed 40 MHz/second and of a power level specified in Appendix I shall cause the fuze transmitter to toggle from f_1 to f_2.

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TABLE I. Arm time.

Arm time setting (seconds)	Arm pulse time (seconds)
1.2	1.2 +0.6, -0.1
4.0	4.0 +0.5, -0.4
6.0	6.0 ±0.6
8.0	8.0 ±0.8
10.0	10.0 ±1.0

TABLE II. HOF setting and accuracies.

HOF switch setting	Selected HOF (feet)	Accuracy (feet)
3	300	±50
5	500	±50
7	700	±50
9	900	+75 -225
12	1,200	+75 -225
15	1,500	+75 -225
18	1,800	+100 -250
22	2,200	+100 -250
26	2,600	+100 -250
30	3,000	+100 -250

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3.2.1.2.1.5.2 ECM/low level immunity. The application of a swept CW jamming signal from f_1 - 10 MHz to f_1 + 10 MHz at a rate not to exceed 10 MHz/second and of a power level specified in Appendix I shall not cause the fuze transmitter to toggle from f_1 to f_2.

3.2.1.2.1.5.3 ECM/spot jammer. The application of a fixed CW jammer signal with a frequency and power level as specified in Appendix I shall cause the fuze transmitter to toggle from f_1 to f_2.

3.2.1.2.2 Time function mode. When both the primary and option arming wires are pulled, the fuze shall arm and provide the explosive output specified in 3.2.1.1.5 within 1.25 +0.6, -0.1 seconds after initiation (see 3.2.1.1.1).

3.2.2 Environmental conditions.

3.2.2.1 Temperature/humidity/altitude. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to four cycles of an environment with temperature extremes of -65 to 149 degrees Fahrenheit (°F), relative humidity to 95 percent and altitudes to 50,000 ft.

3.2.2.2 Salt fog. After exposure to a salt fog environment of 5 percent salt concentration at 95°F, the fuze shall be operable (see 6.3.4) after 48 hours and safe (see 6.3.3) after 96 hours.

3.2.2.3 Extreme temperature storage. None of the safety features of the fuze shall be compromised by exposure, and no components or materials shall in any way be deformed, changed or otherwise altered in such a manner as to degrade performance. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after being exposed to -65°F for 28 days followed by exposure to 160°F for 28 days.

3.2.2.4 Transportation vibration. The fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after vibration testing in all 3 axes at ambient temperature, -65°F, and 160°F, at the following levels.

Frequency (Hz)	Test level
5-11	0.4 in displacement
11-37	2.5 gravity units (g) acceleration, peak
37-52	0.036 in. displacement
52-500	5.0 g acceleration, peak

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3.2.2.5 Random vibration. The fuze shall withstand vibration encountered under captive-flight and free-flight conditions as specified below. None of the safety features of the fuze shall be compromised by exposure, and the fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environments (a) and (b) below.

a. Captive flight, sustained carriage (no buffeting). During captive flight, sustained carriage, the fuze shall withstand vibration at temperatures from -40 to 160°F for 60 minutes along each of the three orthogonal axes (fuze longitudinal, vertical, and aircraft/store level-normal flight attitude), and vibration spectrum (acceleration power spectral density) as follows:

(1) Beginning at 0.01 gravity units squared (g^2)/hertz (Hz) at 20 Hz;

(2) Ramp-up at +3 dB/octave, 20 to 100 Hz.

(3) 0.05 g^2 Hz from 100 to 1,000 Hz; and

(4) Roll-off at -3 dB/octave from 1,000 to 2,000 Hz.

b. Captive flight, attack maneuvering (buffeting). During captive flight, attack maneuvering, the fuze shall withstand vibration at temperatures from -40 to 160°F for a total of 150 seconds along each of the three orthogonal axes (fuze longitudinal, aircraft/store vertical, and aircraft/store level-normal flight attitude), and vibration spectrum (acceleration power spectral density) as follows:

(1) Beginning at 0.5 g^2/Hz at 10 Hz;

(2) Ending at 0.11 g^2/Hz at 100 Hz; and

c. Free flight. During free flight, the fuze shall withstand vibration at temperatures from -40 to 160°F for 6 minutes along each of the three orthogonal axes (fuze longitudinal, aircraft/store vertical, and aircraft/store level - normal flight attitude), and vibration spectrum (acceleration power spectral density) as follows:

20 to 100 Hz	+3 db/octave to 0.05 g^2/Hz
100 to 1000 Hz	0.05 g^2/Hz
1000 to 2000 Hz	-3 db/octave from 0.05 g^2/Hz

3.2.2.6 Temperature shock. The fuze shall withstand exposure to temperatures from -65 to +160°F. None of the safety features of the fuze shall be compromised by exposure, and the fuze shall remain safe and operable.

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3.2.2.7 High temperature. The fuze shall withstand exposure to temperatures up to 160°F followed by a sharp temperature impulse to 220°F for 10 minutes to simulate aerodynamic heating. None of the safety features of the fuze shall be compromised by exposure, and the fuze shall be safe and operable during exposure to each temperature.

3.2.2.8 Low temperature. The fuze shall withstand exposure to temperatures down to -65°F. None of the safety features of the fuze shall be compromised by exposure, and the fuze shall be safe and operable after exposure.

3.2.2.9 Dust. The fuze shall withstand the dust environment for a minimum total time of 28 hours. The dust environment consists of temperature extremes from ambient to 145°F, 1750 feet per minute air velocity and 0.3 grams dust per cubic feet air. None of the safety features of the fuze shall be compromised by exposure, and the fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the environment.

3.2.2.10 Fungus. The fuze shall not support fungus growth when exposed to United States Department of Agriculture (USDA) Fungus Numbers QM 380, 386, 432, 459, and 474. None of the safety features of the fuze shall be compromised by exposure, and the fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to the fungus environment for 28 days.

3.2.2.11 Catapult and arrested landing. The fuze shall withstand the shock from not less than three catapulted aircraft takeoffs and three aircraft arrested landings. With the nose of the fuze oriented as it would be in an ejection-rack-mounted weapon in the direction of aircraft travel, the maximum expected shocks, including a dynamic load factor of 2.0, shall be as follows:

a.	Catapult (takeoff):	4 g forward
		18 g aft
		4 g up
		10 g down
b.	Arrested landing:	18 g forward
		4 g aft
		8 g up
		18 g down

None of the safety features of the fuze shall be compromised by exposure, and the fuze shall be safe (see 6.3.3) and operable (see 6.3.4) after exposure to three shocks in each direction.

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3.2.2.12 Environmental safety conditions.

3.2.2.12.1 Static detonator safety. There shall be no evidence of detonation, burning, charring, scorching, or melting of the explosive lead should the fuze detonator inadvertently function while the fuze is in a safe position (see 6.3.5). There shall be no evidence of rupturing or shattering of the fuze housing or radome, or ejection of particles during or after exposure to the environment of MIL-STD-331, Test 115.

3.2.2.12.2 Jolt. The fuze explosive train shall withstand the jolt from a 4-inch free drop in any orientation onto a padded anvil. No explosive elements shall initiate and the safety features of the fuze shall not be compromised by repeated jolting. The fuze shall be safe to store, transport, and handle, but need not be operable after exposure.

3.2.2.12.3 1.5-meter drop. The fuze shall withstand one 1.5-meter drop in any orientation onto a steel plate. None of the safety features of the fuze shall be compromised by exposure, and the fuze shall be safe to handle and dispose of but need not be operable after exposure.

3.2.2.12.4 12-meter drop. The fuze need not be operable, but shall be safe to handle and dispose of after being dropped from a height of 12 meters onto a steel plate. The fuze lead charge shall not be burned or initiated after exposure.

3.2.2.13 Flight test. The fuze shall meet the performance requirements specified in the flight test plan during exposure to actual flight environments of 4.6.4.13 (see 4.2).

3.2.2.14 Environmental stress screening (ESS). The fuze shall be safe and operable (see 6.3.3 and 6.3.4) after being subjected to the following environmental conditions.

3.2.2.14.1 Temperature Cycling. The fuze shall withstand exposure to six temperature cycles between -40 and 140°F. The cycle time shall be limited to 2 hours at each temperature.

3.2.2.14.2 Vibration. The fuze shall be subjected to 10 minutes of random vibration in the axis shown in Figure 1 and with the power spectral density shown in Figure 2.

3.3 Design and construction.

3.3.1 Production drawings. The fuze shall meet all requirements of and shall be manufactured and assembled in accordance with Drawing 1472AS200 and all associated drawings and documents.

3.3.2 Standards of manufacture.

3.3.2.1 Working environment and product cleanliness. The working environment shall meet the minimum levels and requirements of MIL-STD-1695.

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3.3.2.2 Soldering. Procedures and requirements for preparation and soldering of electrical connections shall be in accordance with WS-6536, unless otherwise specified on the individual drawings associated with Drawing 1472AS200.

3.4 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2 and 6.2.3).

3.4.1 First article sample. The first article sample shall be composed of 243 fully loaded end-item fuzes fabricated in accordance with 3.3.1.

3.5 Workmanship. Fuze workmanship shall meet the requirements of MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspections. Unless otherwise specified in the contract or purchase order (see 6.2.1), the contractor is responsible for the performance of all inspection requirements as specified herein, and the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the contracting activity. The contracting activity reserves the right to perform or witness any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the contracting activity for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the contracting activity to acceptance of defective material.

4.1.1.1 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspections (see 4.2).

b. Quality conformance inspections (see 4.3)

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4.2 First article inspections. First article inspections (see Table III) shall be performed by the contractor and the contracting activity after award of contract and prior to production. First article inspections shall be performed on fuzes which have been produced with the equipment, materials, and procedures to be used in production. First article approval is valid only on the contract under which it is granted unless extended by the contracting activity to other contracts. The contracting activity reserves the right to require additional first article inspections until first article is approved and whenever regular production is interrupted for 120 days or more. Any production prior to approval of the first article shall be at the risk of the contractor.

4.2.1 First article inspection sequence. Prior to first article inspections, all fuzes shall have successfully completed the individual unit acceptance inspections of Figure 3 (see 4.3.1). The first article shall be subjected to the inspections of Figure 4 in the sequence shown. Inspections of Figure 4 shall be performed by the contracting activity or by the contractor (see 6.2.3):

a. Laboratory tests: 39 fully loaded fuzes shall be shipped to the applicable test facility specified in the contract or purchase order for laboratory tests in the sequence shown on Figure 4.

b. Flight tests: 204 fully loaded fuzes shall be shipped to the contracting activity test facility specified in the contract or purchase order for flight tests (see 4.6.4.13).

4.2.2 First article acceptance and rejection criteria. Each first article fuze shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 5, acceptance or rejection of the first article shall be based on the following criteria:

a. For flight tests, accept/reject criteria shall be in accordance with Figure 5.

b. For laboratory tests, defects in the fuze are classified as CRITICAL or MAJOR, as defined in MIL-STD-105, and are specified herein; the first article shall be rejected if:

(1) One or more CRITICAL defects are found, or

(2) Six or more MAJOR defects of unidentical causes are found, or

(3) Two or more MAJOR defects of identical causes are found.

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TABLE III. Inspection/test matrix.

Paragraph number	Requirement	Test paragraph number[1/]	QCI	100% acceptance test	Lot acceptance test	First article test
3.	REQUIREMENTS	-	-	-	-	-
3.1	Item description	-	-	-	-	-
3.2	Characteristics	-	-	-	-	-
3.2.1	Performance	-	-	-	-	-
3.2.1.1	Mechanical	-	-	-	-	-
3.2.1.1.1	Fuze initiation	4.6.2.1	X	-	X	X
3.2.1.1.2	Time function mode select	4.6.2.2	X	-	X	X
3.2.1.1.3	Air velocity sensor	4.6.2.3	X	X	X	X
3.2.1.1.4	Seal integrity	4.6.2.4	X	X	X	X
3.2.1.1.5	Explosive output	4.6.2.5	X	-	X	X
3.2.1.1.6	Operating life	4.6.2.6	X	-	X	X
3.2.1.2	Electrical	-	-	-	-	-
3.2.1.2.1	Proximity function mode	4.6.3.1.1	X	-	X	X
3.2.1.2.1.1	Arm time	4.6.3.1.2	X	-	X	X

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TABLE III. Inspection/test matrix. (Continued)

Paragraph number	Requirement	Test paragraph number 1/	QCI	100% acceptance test	Lot acceptance test	First article test
3.2.1.2.1.2	Height of function (HOF)	-	-	-	-	-
3.2.1.2.1.2.1	Primary HOF	4.6.3.1.3	X	X	X	X
3.2.1.2.1.2.2	Alternate HOF	4.6.2.1.4	X	X	X	X
3.2.1.2.1.3	Loop loss sensitivity	4.6.3.1.5	X	X	X	X
3.2.1.2.1.4	Operating frequency	4.6.3.1.6	X	X	X	X
3.2.1.2.1.5	ECM immunity	-	-	-	-	-
3.2.1.2.1.5.1	ECM high level immunity	4.6.3.1.7	X	X	X	X
3.2.1.2.1.5.2	ECM low level immunity	4.6.3.1.8	X	X	X	X
3.2.1.2.1.5.3	ECM spot jammer	4.6.3.1.9	X	X	X	X
3.2.1.2.2	Time function mode	4.6.3.1.10	X	X	X	X
3.2.2	Environmental conditions	-	-	-	-	-
3.2.2.1	Temperature-humidity-altitude	4.6.4.1	X	-	-	-
3.2.2.2	Salt fog	4.6.4.2	X	-	-	X

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TABLE III. Inspection/test matrix. (Continued)

Paragraph number	Requirement	Test paragraph number[1/]	QCI	100% acceptance test	Lot acceptance test	First article test
3.2.2.3	Extreme temperature storage	4.6.4.3	X	-	-	X
3.2.2.4	Transportation vibration	4.6.4.4	X	-	-	X
3.2.2.5	Random vibration	4.6.4.5	X	-	X	X
3.2.2.6	Temperature shock	4.6.4.6	X	-	-	X
3.2.2.7	High temperature	4.6.4.7	X	-	X	X
3.2.2.8	Low temperature	4.6.4.8	X	-	X	X
3.2.2.9	Dust (fine sand)	4.6.4.9	X	-	-	X
3.2.2.10	Fungus	4.6.4.10	X	-	-	X
3.2.2.11	Catapult and arrested landing	4.6.4.11	X	-	-	X
3.2.2.12	Environmental safety conditions	-	-	-	-	-
3.2.2.12.1	Static detonator safety	4.6.4.12.1	X	-	-	X
3.2.2.12.2	Jolt	4.6.4.12.2	X	-	-	X
3.2.2.12.3	1.5-meter drop	4.6.4.12.3	X	-	-	X

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TABLE III. Inspection/test matrix. (Continued)

Paragraph number	Requirement	Test paragraph number 1/	QCI	100% acceptance test	Lot acceptance test	First article test
3.2.2.12.4	12-meter drop	4.6.4.12.4	X	-	-	X
3.2.2.13	Flight test	4.6.4.13	X	-	-	X
3.2.2.14	Environmental stress screening	-	-	-	-	-
3.2.2.14.1	Temperature cycling	4.6.4.14.1	X	X	-	-
3.2.2.14.2	Vibration	4.6.4.14.2	X	X	-	-
3.3	Design and construction	-	-	-	-	-
3.3.1	Production drawings	4.6.1a	X	X	X	X
3.3.2	Standards of manufacture	4.6.1a thru d	X	X	X	X
3.4	First article	4.2	-	-	X	
3.5	Workmanship	4.6.1a thru d	X	X	X	X
5.	PACKAGING	4.6.5	X	X	X	X

1/ 4.6.3.1 Electronic diagnostic tests applies from 3.2.1.2 to 3.3.

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NOTE: Although five MAJOR defects do not cause rejection of the first article, appropriate corrective action shall be taken for each MAJOR defect found to prevent recurrence prior to further inspection or production. Each defect shall be documented as specified in 4.5d.

c. The first article shall be rejected if, during any flight or laboratory test, the fuze arms early or if any explosive component of the fuze initiates early.

4.3 Quality conformance inspections. Quality conformance inspections shall be performed as specified in Table III to verify the requirements of sections 3 and 5.

4.3.1 Individual unit acceptance inspections. Individual unit acceptance inspections (100 percent) shall be performed in accordance with Figure 3 during manufacture of the fuze and prior to formation of the first article or the inspection lot. Fuzes found to be defective during the inspection sequence shall be rejected on an individual basis.

4.3.2 Lot inspection.

4.3.2.1 Lot size. The lot size shall consist of not less than 50 or more than 280 fuzes and shall be identified by assignment of a lot number.

4.3.2.2 Lot inspection sequence. Prior to formation of the lot, all fuzes shall have passed the individual unit acceptance inspections of Figure 3 (see 4.3.1). The required quantity of sample fuzes (see 4.3.2.2.1) shall then be randomly selected from the lot. The sample fuzes shall be subjected to the applicable inspections of Figure 6 in the sequence shown; the remainder of the lot shall be held in bond pending successful completion of the sampling inspection sequence. If the sample fuzes do not cause rejection, the remainder of the fuzes shall then be removed from bond.

4.3.2.2.1 Sampling plan. Sampling shall be as specified below.

TABLE IV. Sampling plan.

Fuze lot size	Fuze sample size	Accept	Reject
50 - 100	5	0	1
101 - 150	10	1	2
151 - 280	15	1	2

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4.3.2.3 Lot acceptance and rejection criteria. Each fuze shall meet all the requirements specified in the contract or purchase order, on the drawings, and herein. For the inspections of Figure 6, acceptance or rejection of the fuzes shall be based on the following criteria:

a. Where 100-percent inspection is required, acceptance or rejection shall be on an individual basis.

b. For inspections on sample fuzes, defects in the fuze are classified as CRITICAL or MAJOR, as defined in MIL-STD-105. The fuzes shall be rejected if one or more CRITICAL defects are found. For MAJOR defects, acceptance and rejection shall be in accordance with Table IV.

c. The sampling inspection lot shall be rejected if, during any test, the fuze arms early or if any explosive component of the fuze initiates early.

4.4 Measuring and test equipment.

4.4.1 Calibration and maintenance. All measuring and test equipment used by the contractor for performance of the inspections specified herein shall be calibrated and maintained in accordance with MIL-STD-45662.

4.4.2 Standard test equipment. The contractor shall provide and maintain inspection and test equipment of sufficient quality and quantity to permit the performance of the required tests and measurements, under the conditions, and to the accuracies specified herein. Unless otherwise specified herein, the magnitude of any error introduced by the test equipment shall not exceed 10 percent of the allowable tolerance of the requirement being measured. Any error introduced by the test equipment shall not be applicable toward widening the allowable tolerance of the requirement being measured.

4.4.3 Special test equipment and circuits. When special test equipment and circuits, other than as specified herein or supplied by the Government, are devised or commercially available equipment is employed, all test equipment, circuits, and methods shall meet the requirements of 4.4.2 and are subject to verification by the contracting activity.

4.4.4 Gauges. The contractor shall provide the gauges necessary to ensure conformance to the dimensional requirements of this specification and the applicable drawings. The contracting activity will furnish drawings of pertinent Government final inspection gauges (see 6.2.1). This procedure does not relieve the contractor of his responsibility for the design and manufacture of the gauges required for fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Government final acceptance gauges.

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4.5 Test conditions and provisions. Unless otherwise specified herein, the following test conditions and provisions shall apply for performance of the inspection methods of 4.6.

a. Safety precautions. The fully-loaded fuze contains explosives which are specified on Drawing 1472AS200 and associated drawings. The inspections of 4.6 require the exercise of extremely stringent safety precautions and procedures, experienced ordnance personnel, suitable equipment and facilities, and comprehensive safety reviews (see 6.4).

b. Test environment. Unless otherwise specified herein, the inspection methods of 4.6 shall be performed under the conditions specified in 3.2.2.12.1.

c. Test fixtures. Unless otherwise specified in the contract or purchase order (see 6.2.1), all test fixtures for performance of the inspections of 4.6 shall be supplied by the contractor and shall be verified by the contracting activity prior to use.

d. Shipment lot inspection report. The contractor shall prepare a report for each lot produced under a specified contract, including first article inspection and rejected lots (see 6.2.2). The report shall include the following:

(1) A summary listing of all deviations, waivers and drawings or specifications changes applicable to the lot.

(2) A summary of acceptance test results on the sample fuzes as follows:

(a) Arming time data.

(b) A listing of defects that occurred during the sampling inspection tests according to fuze identification number, defect classification, and drawing number or specification paragraph.

e. Inspection sequences. The inspections of 4.6 shall be performed in the sequences shown on Figures 3, 4, and 6, as applicable. When feasible, sequential inspections may be combined, provided it can be demonstrated by the contractor that the applicable requirements of Section 3 have been individually met.

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4.6 Inspection methods. To determine conformance to the requirements of Section 3, the inspections specified in 4.5 shall be performed in accordance with the test conditions and provisions of 4.6.1 through 4.6.5.

WARNING

The fuze contains explosive components that are sensitive and hazardous. Suitable safety precautions shall be observed during handling and testing of the fuze (see 4.5a and 6.4).

4.6.1 Visual examination. The fuze shall be visually examined as follows:

a. For conformance to Drawing 1472AS200.

b. There shall be no evidence of external damage, such as cracks, dents, or other blemishes, which may interfere with functioning or fit of the fuze at the next assembly.

MAJOR

c. The fuze shall be in the safe position (see 6.3.5).

CRITICAL

d. The warning streamer and lock pin (see Drawing 1472AS200) shall be correctly installed.

MAJOR

4.6.2 Mechanical. To verify conformance to 3.2.1.1 the following operability tests shall be performed.

4.6.2.1 Fuze initiation. Apply a pull force of 125 ±10 pounds to the fuze primary arming wire to verify conformance to 3.2.1.1.1.

4.6.2.2 Time function mode select. To verify conformance to 3.2.1.1.2, apply a pull force of 125 ±10 pounds to the option arming wire in conjunction with fuze initiations (see 4.6.2.1).

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4.6.2.3 Air velocity sensor. The following tests shall be performed to verify conformance to 3.2.1.1.3:

a. Test 1. Apply an air pressure differential to the air velocity sensor that corresponds to 160 +0 -15 KCAS. Accept the fuze if the impedance at the arm pulse output of J1 (see Drawing 1472AS200) is more than 40 ohms.

WARNING

Under no circumstances should power be applied to the fuze before, during, or immediately following this test, since the fuze contains sensitive explosives that can cause personnel injury.

Remain safe: CRITICAL

b. Test 2. Apply an air pressure to the fuze air velocity sensor that corresponds to 225 +25 -0 KCAS. Accept the fuze if the impedance at the arm pulse output of J1 (see Drawing 1472AS200) is less than 15 ohms.

CRITICAL

4.6.2.4 Seal integrity. To verify conformance to 3.2.1.1.4, pressurize the fuze to 7.5 ±0.5 pounds per square inch gauge (psig). Then, completely immerse the pressurized fuze 2 ±1 inches below the surface of an alcohol bath that is temperature-conditioned to room ambient temperature. Visually observe the fuze for any evidence of gas leakage. Accept the fuze if leakage is not present for a period of 30 seconds minimum after immersion of the fuze. Alternate test methods may be used with prior approval by the Government activity.

MAJOR

4.6.2.5 Explosive output. The explosive output of 3.2.1.1.5 shall be verified by initiation of the explosive train with the fuze installed in a test fixture which retains the dent block (see 6.3.7). Verification of the output shall be made by measuring the depth of the dent (see 3.2.1.1.5).

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4.6.2.6 Operating life. Verify conformance to 3.2.1.1.6. Initiate the fuze in accordance with 3.2.1.1.1 and monitor the battery voltage or fuze operating frequencies.

MAJOR

4.6.3 Electrical.

4.6.3.1 Electronic diagnostic tests (EDT). These tests shall be used to verify that the fuze can meet the requirements of 3.2.1.2 through 3.2.1.2.2. Unless otherwise specified, the following test conditions and provisions shall apply:

a. Since the thermal battery in the fuze is a one-shot, non-cyclical device, fuze initiation (see 3.2.1.1.1) shall be simulated by applying power to J1 (see 1472AS200) of the fuze as follows:

(1) +18 V input of J1 (see Drawing 1472AS200) powered with +18 ±0.5 Vdc.

(2) -18 V input of J1 (see Drawing 1472AS200) powered with -18 ±0.5 Vdc.

b. A detectable target shall be one that delays the transmit signal within the tolerance of the desired HOF settings, as expressed in Table II, modulated with a Doppler signal simulating a vertical velocity of 1,200 fps and retransmits the signal to the fuze with a loop loss of no less than 122 dB, except as noted in 3.2.1.2.1.3.

4.6.3.1.1 Proximity function mode. To verify conformance to 3.2.1.2.1, initiate the fuze (see 3.2.1.1.1). The fuze shall be acceptable if the fuze meets the requirements of 3.2.1.2.1.1 through 3.2.1.2.1.4.

4.6.3.1.2 Arm time. To verify conformance to 3.2.1.2.1.1, verify the occurrence of an arm pulse in the fuze by measuring the time interval from fuze initiation (see 4.6.2.1) to the appearance of a voltage of more than +6 Vdc at the arm pulse output of J1 (see Drawing 1472AS200). Failure to meet the above requirements shall be cause for rejection.

MAJOR

4.6.3.1.3 Primary HOF. To verify conformance to 3.2.1.2.1.2.1, verify the 10 primary HOF settings of the fuze by initiating the fuze (see 4.6.2.1) and monitoring the fuze range gates on an oscilloscope as the HOF switch is rotated through its 10 positions. The fuze shall be acceptable if the range gate pulse for each HOF setting exceeds 10 millivolts and the spurious or undesired range gates are under 5 millivolts.

MAJOR

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4.6.3.1.4 Alternate HOF. To verify conformance to 3.2.1.2.1.2.2, monitor the oscilloscope DISPLAY during the performance of 4.6.3.1.3. The fuze shall be acceptable if the alternate HOF range gate is observed for HOF switch settings 5 through 30.

MAJOR

4.6.3.1.5 Loop loss sensitivity. To verify conformance to 3.2.1.2.1.3, initiate the fuze (see 4.6.2.1) and provide the fuze with a detectable target (see 4.6.2.1) that has a vertical closing velocity and loop loss as specified in 3.2.1.2.1.3 and a HOF switch setting of 900 feet. The fuze shall be acceptable if the fuze provides a fire pulse at the fire pulse output of J1 (see Drawing 1472AS200).

MAJOR

4.6.3.1.6 Operating frequency. To verify conformance to 3.2.1.2.1.4, monitor the operating frequencies of the fuze during ECM testing. (see 3.2.1.2.1.5).

MAJOR

4.6.3.1.7 Swept high CW jamming. To verify conformance to 3.2.1.2.1.5.1, initiate the fuze (see 4.6.2.1) and apply CW jamming to the fuze as specified in 3.2.1.2.1.5.1. The fuze shall be acceptable if the fuze toggles from f_1 to f_2.

MAJOR

4.6.3.1.8 Swept low CW jamming. To verify conformance to 3.2.1.2.1.5.2, initiate the fuze (see 4.6.2.1) and apply a swept CW jamming signal to the fuze as specified in 3.2.1.2.1.5.2. The fuze shall be acceptable if the fuze does not toggle from f_1 to f_2.

MAJOR

4.6.3.1.9 ECM/spot jammer. Apply a fixed CW jamming signal with a frequency and power level selected from Appendix I and verify that the fuze transmitter toggles in accordance with 3.2.1.2.1.5.3.

MAJOR

4.6.3.1.10 Time function mode. To verify conformance to 3.2.1.2.2, apply an axial pull force of 125 ±10 pounds to the option arming wire and initiate the fuze in accordance with 3.2.1.1.1. If the output of 3.2.1.2.2 occurs within the tolerance specified, the fuze shall be acceptable.

MAJOR

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4.6.4 Environmental tests.

4.6.4.1 Temperature/humidity/altitude. To verify conformance to 3.2.2.1, test the fuze in accordance with the following:

Step 1. Place the fuze in the test chamber.

Step 2. Reduce the chamber temperature to -65°F within 2 hours.

Step 3. Reduce the chamber pressure at a rate of 1000 to 1500 ft/min to 50,000 feet altitude while maintaining -65°F. The time from start to finish of step 2 shall be 2-1/2 hours.

Step 4. Within 30 minutes, return the chamber temperature and pressure to standard ambient and then raise the chamber humidity to 95 percent relative humidity (RH).

Step 5. Maintain chamber conditions for 2-1/2 hours.

Step 6. During the next 30 minutes, increase the temperature to 149°F while maintaining humidity at 95 percent RH.

Step 7. Maintain chamber conditions for 6 hours.

Step 8. During the next 8 hours, reduce the chamber temperature at a uniform rate to standard ambient while maintaining 95 percent RH.

Step 9. Maintain chamber conditions for 2 hours.

Steps 2 through 9 constitute one cycle. This cycle shall be performed a total of four times. After the four cycles are completed, the following shall be performed:

a. Repeat step 2

b. Repeat step 3

c. Within 30 minutes, return the chamber to standard ambient

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.2 Salt fog. To verify conformance to 3.2.2.2, test the fuze in accordance with MIL-STD-331, Test 107.1.

Remain safe: CRITICAL
Remain operable: MAJOR

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4.6.4.3 Extreme temperature storage. To verify conformance to 3.2.2.3, test the fuze in accordance with MIL-STD-331, Test 112.1, Procedure I.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.4 Transportation vibration-temperature. To verify conformance to 3.2.2.4, perform the following tests:

4.6.4.4.1 Transportation vibration-ambient. Ensure that the fuze is tested in accordance with MIL-STD-331, Test 119, Procedure I, with 5.1.2 modified to exclude the 160°F and -65°F tests (b) and (c) conditions of the first sentence. Vibration (5-500 Hz) shall be performed for 4 hours per axis, for a total of 12 hours.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.4.2 Transportation vibration-hot. Test the fuze in accordance with MIL-STD-331, Test 119, Procedure I, with 5.1.2 modified to exclude the ambient and -65°F tests (a) and (c) conditions of the first sentence. Vibration (5-500 Hz) shall be performed for 4 hours per axis, for a total of 12 hours.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.4.3 Transportation vibration-cold. Test the fuze in accordance with MIL-STD-331, Test 119, Procedure I, with 5.1.2 modified to exclude the ambient and 160°F tests (a) and (b) conditions of the first sentence. Vibration (5-500 Hz) shall be performed for 4 hours per axis, for a total of 12 hours.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.5 Random vibration. To verify conformance to 3.2.2.5, test the fuze in accordance with MIL-STD-810, Method 514.3, Procedure I, with vibration inputs and levels as follows:

a. For 3.2.2.5a, the vibration spectrum shall be in accordance with Figure 7. The test time shall be not less than 1 hour for each axis, and the test temperature shall be laboratory ambient.

Remain safe: CRITICAL
Remain operable: MAJOR

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b. For 3.2.2.5b, the vibration spectrum shall be in accordance with Figure 8. The test time shall be 150 seconds per axis, and the test temperature shall be -40 F.

Remain safe: CRITICAL
Remain operable: MAJOR

c. For 3.2.2.5c, the spectral shape shall be in accordance with Figure 7. Test duration shall be 6 minutes per axis. Test article temperature shall be laboratory ambient.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.6 Temperature shock. To verify conformance to 3.2.2.6, test the fuze in accordance with MIL-STD-810, Method 503.2, Procedure I. The temperature in step 1 shall be -65°F and the temperature in step 2 shall be 160°F. Temperature stabilization times shall be 3 hours (steps 1 and 2).

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.7 High temperature. To verify conformance to 3.2.2.7, test the fuze in accordance with MIL-STD-810, Method 501.2, Procedure II for constant temperature exposure (not cyclic). The test shall be conducted at 160 ±5°F for 2 +1 -0 hours then the temperature shall be raised to 220 ±5°F for 10 ±1 minutes. Operate the fuze within 2 minutes after removal from the temperature chamber.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.8 Low temperature. To verify conformance to 3.2.2.8, test the fuze in accordance with MIL-STD-810, Method 502.2, Procedure II. The temperature shall be -65 ±5°F for 2 hours minimum. Operate the fuze within 2 minutes after removal from the temperature chamber.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.9 Dust (fine sand). To verify conformance to 3.2.2, test the fuze in accordance with MIL-STD-331, Test 116.1.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.10 Fungus. To verify conformance to 3.2.2.10, test the fuze in accordance with MIL-STD-331, Test 110.1, all phases, test item configurations (1) and (2).

Remain safe: CRITICAL
Remain operable: MAJOR

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4.6.4.11 Catapult and arrested landing. To verify conformance to 3.2.2.11, test the fuze in a suitable test fixture in accordance with MIL-STD-331, Test 212.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.4.12 Environmental safety conditions.

4.6.4.12.1 Static detonator safety. To verify conformance to 3.2.2.12.1, test the fuze in accordance with MIL-STD-331, Test 115.2.

Remain safe: CRITICAL

4.6.4.12.2 Jolt. To verify conformance to 3.2.2.12.2, test the fuze in accordance with MIL-STD-331, Test 101.3, Procedure II.

Remain safe: CRITICAL

4.6.4.12.3 1.5-meter drop. To verify conformance to 3.2.2.12.3, test the fuze in accordance with MIL-STD-331, Test 111.1, Procedure II, Criteria A, drop orientations as follows:

a. Along the longitudinal axis (coincides with longitudinal axis of the weapon).

Remain safe: CRITICAL

b. Along a line through the center of rotation of the fuze explosive train interrupter.

Remain safe: CRITICAL

4.6.4.12.4 12-meter drop. To verify conformance to 3.2.2.12.4, test the fuze in accordance with MIL-STD-331, Test 103.2, Procedure III.

Remain safe: CRITICAL

4.6.4.13 Flight test. To verify conformance to 3.2.2.13 and when specified in the contract or purchase order, the Government will perform flight tests on a sample quantity of fuzes from the first article in accordance with the test conditions specified in the test plan.

4.6.4.14 Environmental stress screening.

4.6.4.14.1 Temperature cycling. To verify conformance to 3.2.2.14.1 subject the fuze to six hot/cold temperature cycles. A cycle shall be 2 hours at -40°F followed by 2 hours at 140°F.

Remain safe: CRITICAL
Remain operable: MAJOR

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4.6.4.14.2 Vibration. To verify conformance to 3.2.2.14.2 expose the fuze to 10 minutes of random vibration along the axis indicated on Figure 1. The vibration power spectral density shall be as indicated on Figure 2.

Remain safe: CRITICAL
Remain operable: MAJOR

4.6.5 Packaging. To determine conformance to Section 5, visually inspect the packaging, packing and marking of the fuze prior to shipping.

5. PACKAGING

5.1 Preservation. Preservation shall be in accordance with FED-STD-102, Level A, and as specified on Drawing 1472AS400. Preservation shall conform to 49 CFR 171-178.

5.2 Packing. Packing shall be in accordance with FED-STD-102, Level A, and as specified on Drawing 1472AS400. The shipping container has space for two fuzes. Fuzes, packaged in accordance with 5.1, shall be packed in accordance with Drawing 1472AS400. Packing and containers shall be suitable for explosives and shall conform to 49 CFR 171-178. Navy palletizing plan shall be in accordance with MIL-STD-1322.

5.2.1 Navy ammunition data card. One Navy ammunition data card, completed in accordance with MIL-STD-1167, and listing the explosive components, shall be packed inside each shipping container.

5.3 Marking. In addition to any special marking required by the contract or purchase order, all interior and exterior marking shall be in accordance with MIL-STD-129 including bar coding and Drawing 1472AS400 (see 6.2.1).

6. NOTES

6.1 Intended use. The fuze is intended for use on Rockeye II (MARK 20 MODS 3, 4, 6, and 7), APAM (CBU-59/B), Gator (CBU-78/B), and Bigeye (BLU-80/B) weapon systems.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Responsibility for inspection facilities and approval of test fixtures, if other than as specified in 4.1, and 4.5.c.

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c. Details for flight testing by the Government (see 4.2.1.b).

d. Government final inspection gauge drawings, if required (see 4.4.4).

e. Inspection lot size (see 4.3.2.1).

f. First article sample size (see 4.2.1).

g. Special marking, if required (see 5.3).

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below should be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub.Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below should be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following Paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.	Option
(a) 4.2, 4.3	Test Plan	DI-T-5204	---
(b) 4.4	Certification Data/ Report	UDI-A-23264B	---
(c) 5.2.1	Ammunition Data Card	DI-MISC-80043	---

(Data item descriptions related to this specification, and identified in Section 6, will be approved and listed as such in DOD 5010.12L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.2.3 First article. When a first article inspection is required, the sample fuzes shall be first article units tested and inspected in accordance with 4.2. The first article sample should consist of 243 fully-loaded fuzes, as specified in 4.2.1. The contracting officer should include specific instructions in the acquisition documents for the following:

a. The arrangements for examinations, test approval, and final disposition of the first article test samples.

b. Identification of the test facility to perform the laboratory tests of 4.2.

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c. Identification of the Government test facility to perform flight testing, if required, and applicable details of flight tests, including performance requirements, test conditions, and scheduling.

d. Identification of a facility to perform a physical configuration audit, if required by the contracting activity, and details for the audit.

e. Approval of the first article samples and tests shall be furnished by the contracting officer.

6.3 Definitions.

6.3.1 Initiation. Fuze initiation is defined as that event when the thermal battery is activated by pulling the fuze arming wire.

6.3.2 Loop loss. Loop loss is defined as the attenuation from transmission to reception of the fuze RF signal.

6.3.3 Safe. Safe is defined as the ability of the fuze to maintain its safety features in a condition which will not create a hazard for personnel or cause any subsequent action to occur during its service life that nullifies the required safety conditions of service use. For nonservice use, safe is defined as the ability of the fuze to maintain its safety features in a condition which will permit disposal without injury to personnel.

6.3.4 Operable. Operable is defined as the ability of the fuze to perform to completion its specified function and sequence, and produce the specified outputs within the required operational period, and environment. For the environmental requirements and tests specified herein, operable is further defined as the ability of the fuze to meet the requirements of 3.2.1 after exposure to the environments of 4.3, as required in the individual test method.

6.3.5 Safe position. The safe position is defined as the rotor mechanically locked in the out-of-line position indicated by no evidence of the red color from the fuze indicator rod being visible. The indicator rod is visible if it extends beyond the outer surface of the fuze radome.

6.3.6 Armed position. The armed position is defined as the rotor mechanically locked in the in-line position indicated by the red indicator rod extending beyond the outer surface of the fuze radome.

6.3.7 Dent block. The dent block shall be 1.50 x 1.50 x 0.625 inches (all dimensions ±0.050) steel, cold finished in accordance with ASTM A 108, Grade 1018 or 1020, Rockwell hardness B20 to B95.

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6.4 Explosive safety requirements. Provide minimum explosive safety requirements for use by the contractor, which is provided in DoD instruction 4145.26-M "DoD Contractor's Safety Manual for Ammunition and Explosives (Dated March 86).

6.5 Subject term (key word) listing.

Fuzes (ordnance)
Proximity fuzes
FMU 140/B dispenser proximity fuze
Radar altimeter
Dispenser weapons
Rockeye II
APAM
Gator
Bigeye

Preparing activity
Navy-AS
(Project 1325-N130)

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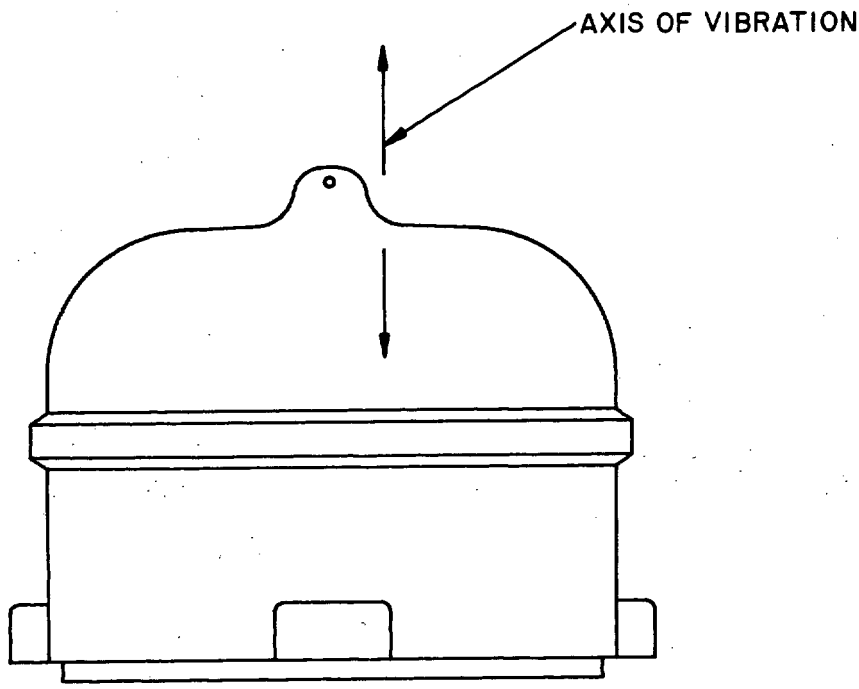


FIGURE 1. Environmental stress screening vibration axis.

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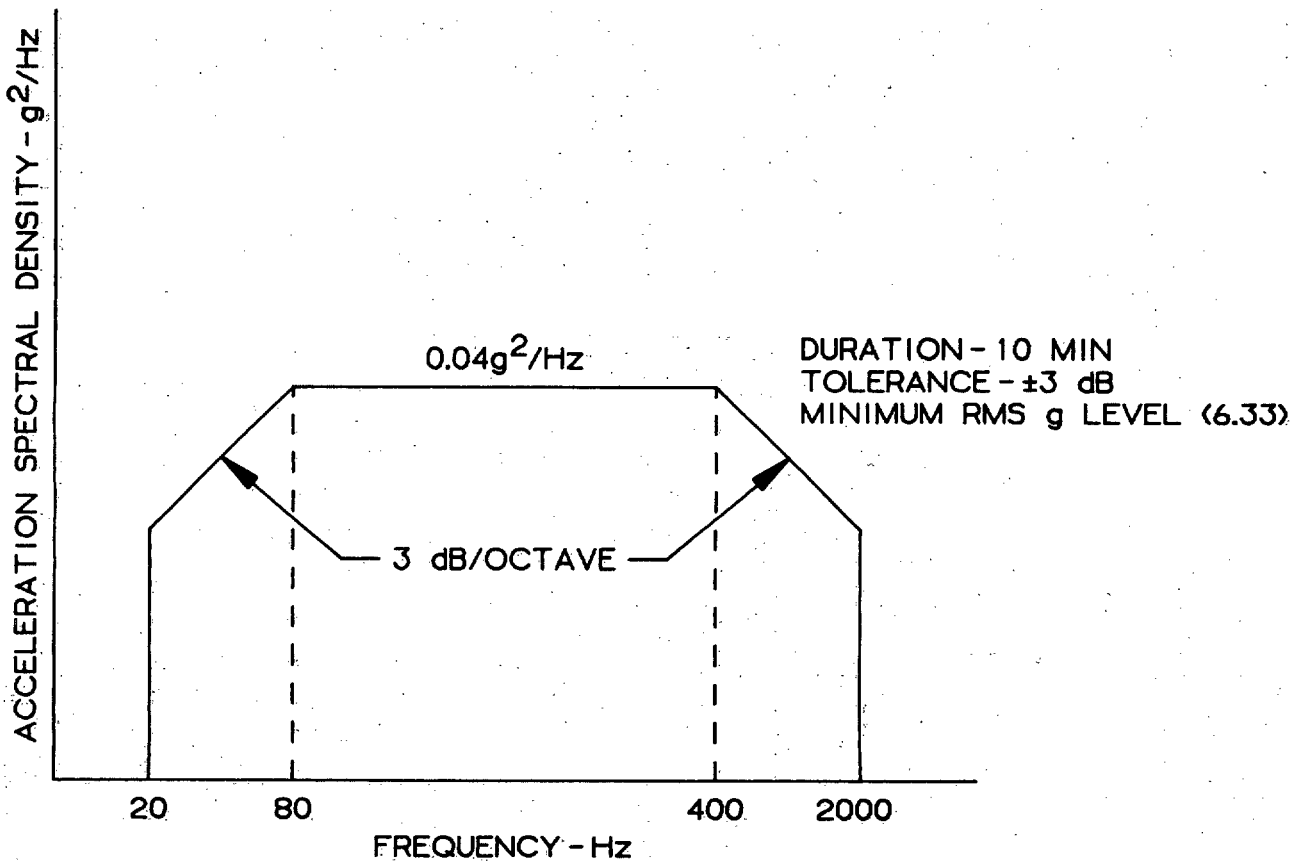


FIGURE 2. ESS vibration power spectral density.

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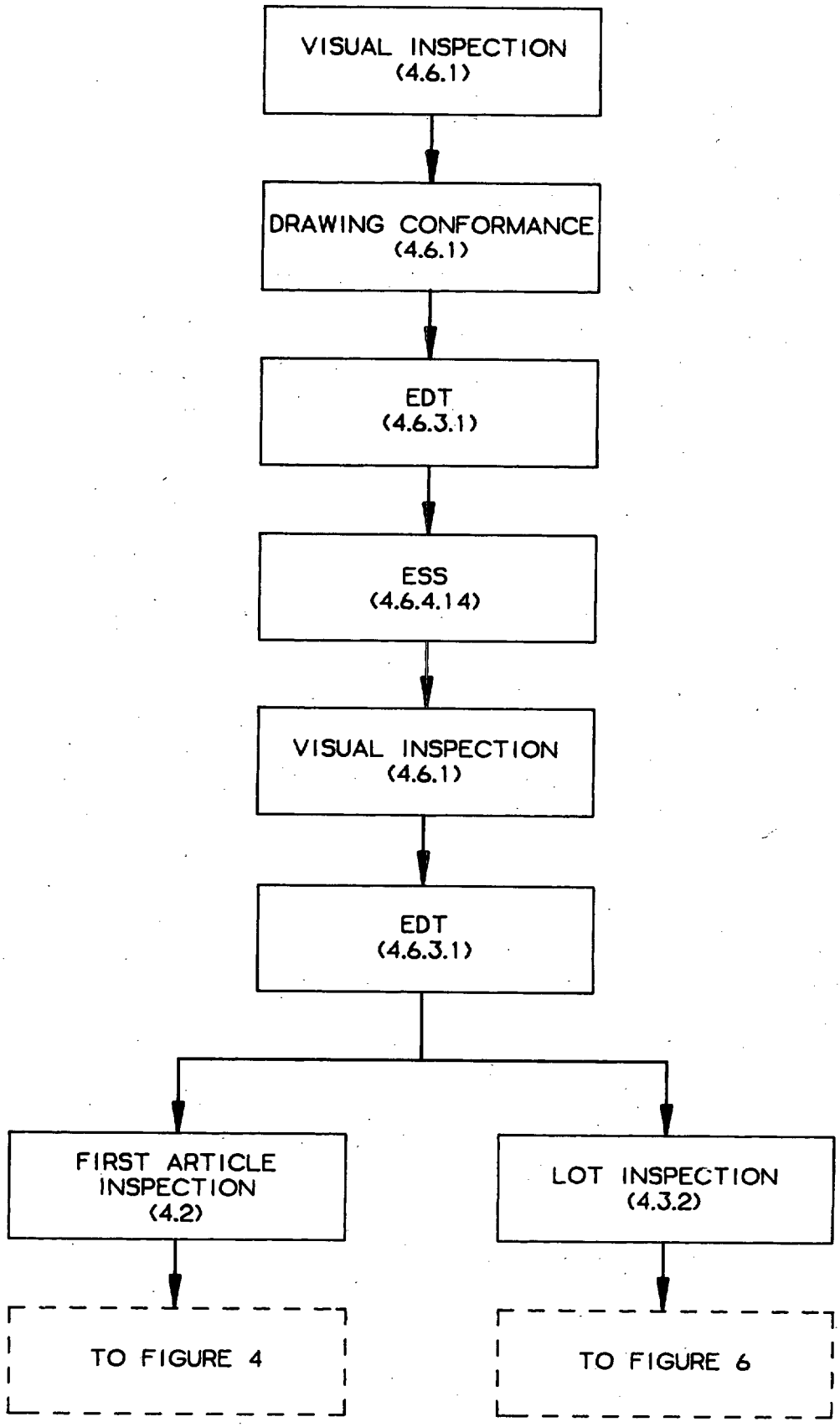
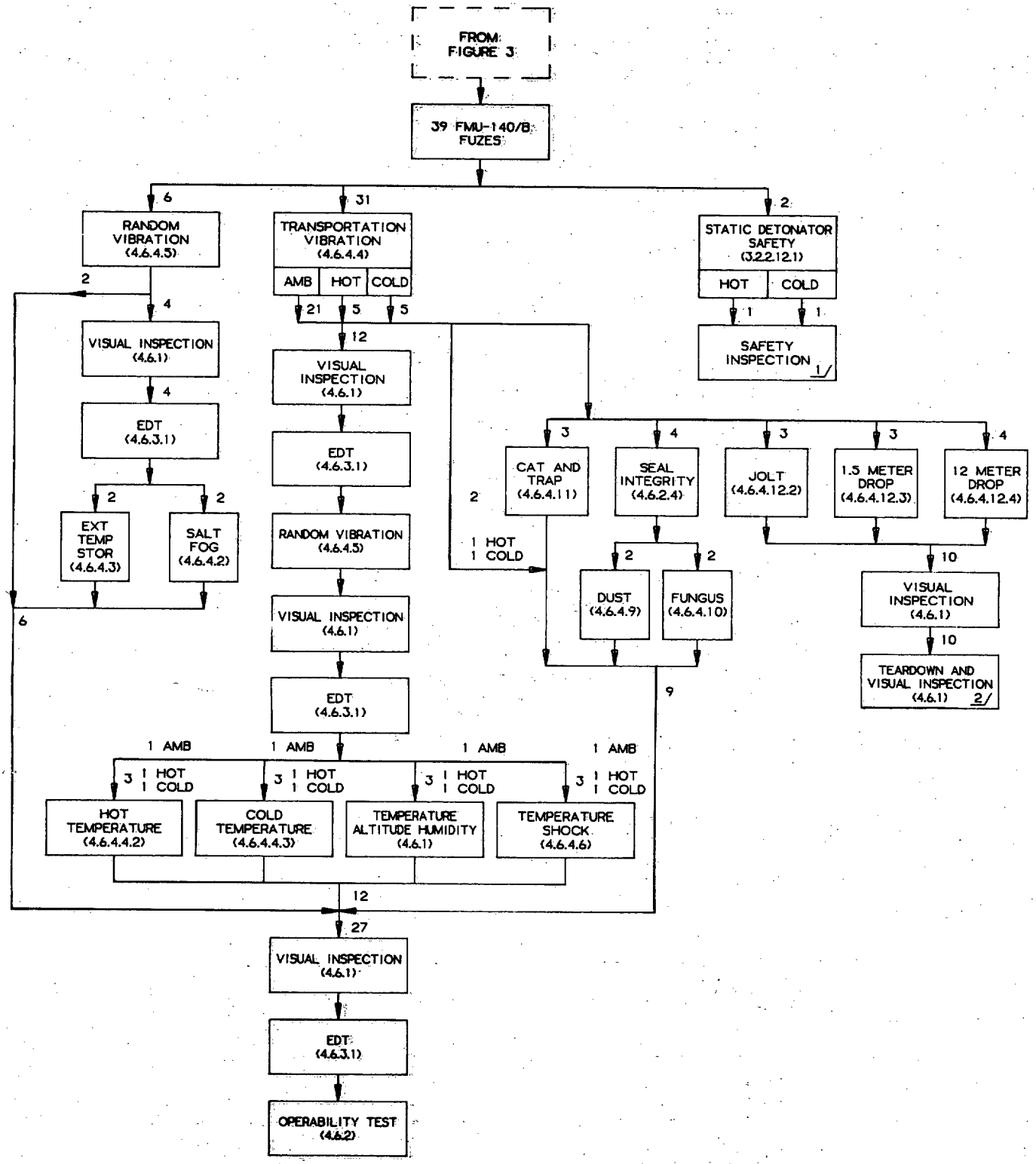


FIGURE 3. Individual unit acceptance inspection sequence.

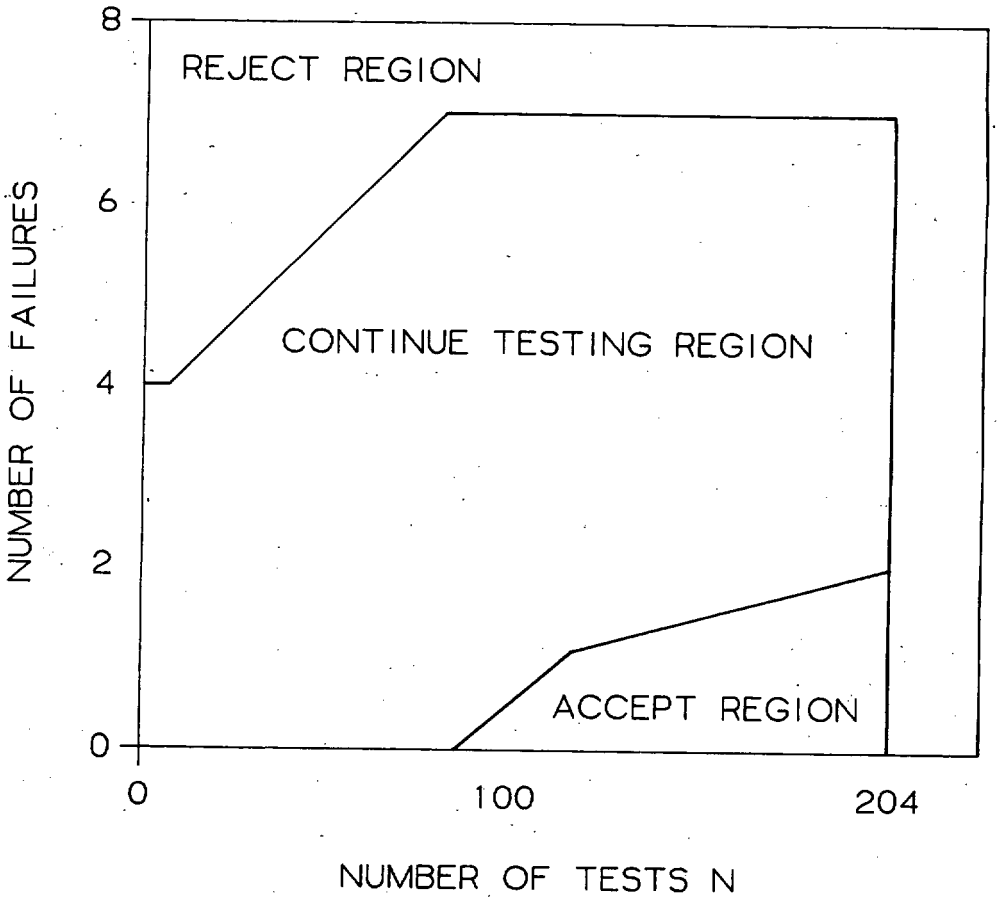
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:52Z
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1/ VERIFY FUZE IS SAFE OR ARMED (SEE 6.3).
2/ TEARDOWN-DISASSEMBLE FUZE AND INSPECT.

FIGURE 4. First article inspection sequence.

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NUMBER OF FAILURES	ACCEPT IF N ≥	REJECT IF N ≤
0	97	0
1	121	0
2	204	0
3	204	0
4	204	0
5	204	25
6	204	51
7	204	76

NOTE: REJECT ON 8 FAILURES REGARDLESS OF VALUE OF N

FIGURE 5. Flight test acceptance and rejection criteria.

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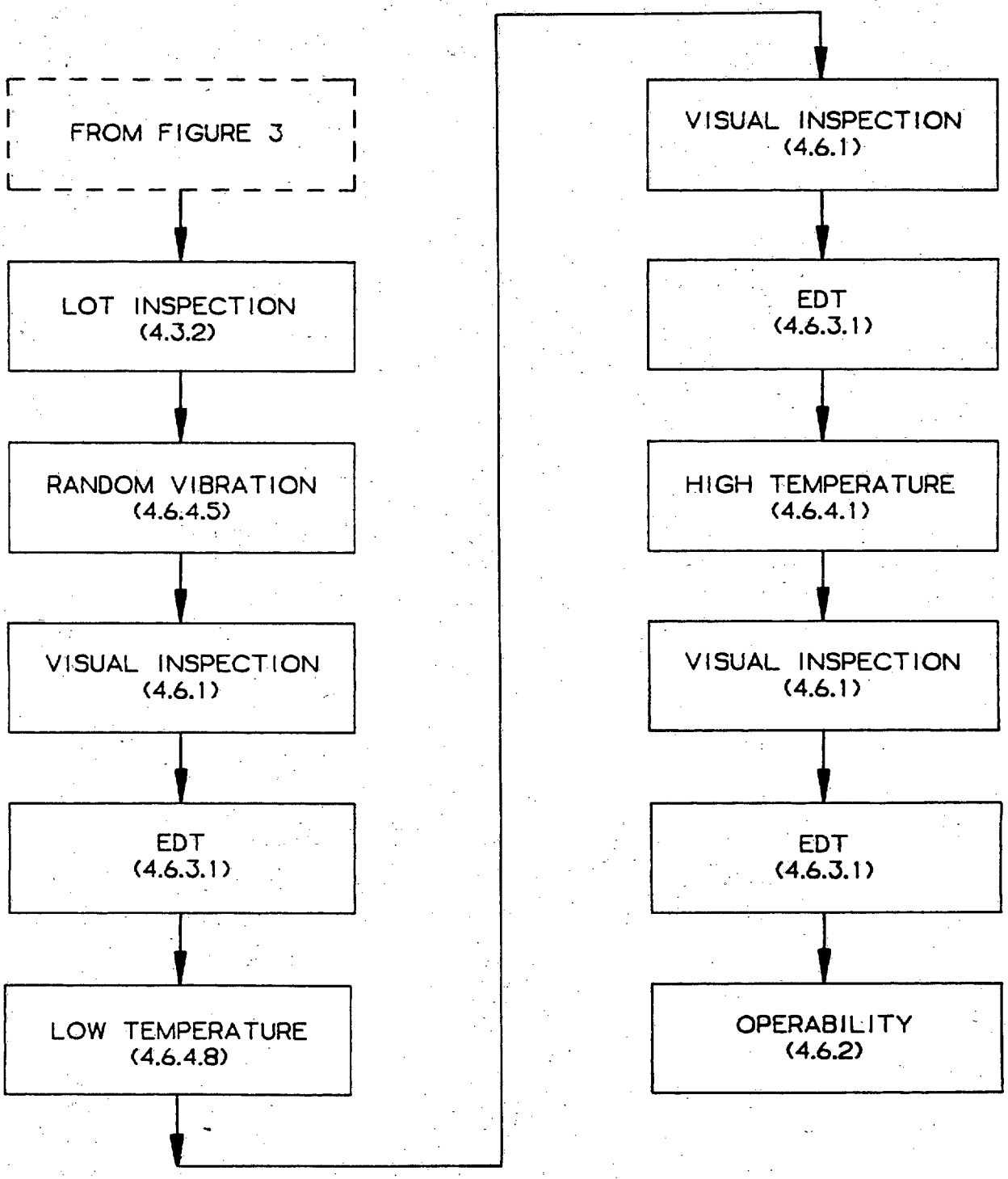


FIGURE 6. Lot inspection sequence.

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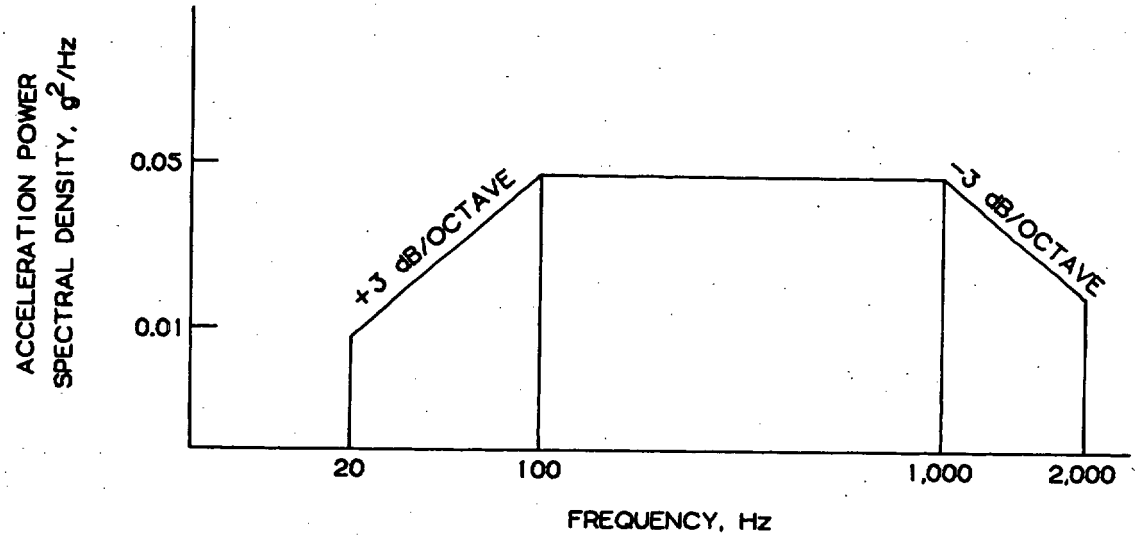


FIGURE 7. Captive flight random vibration spectrum.

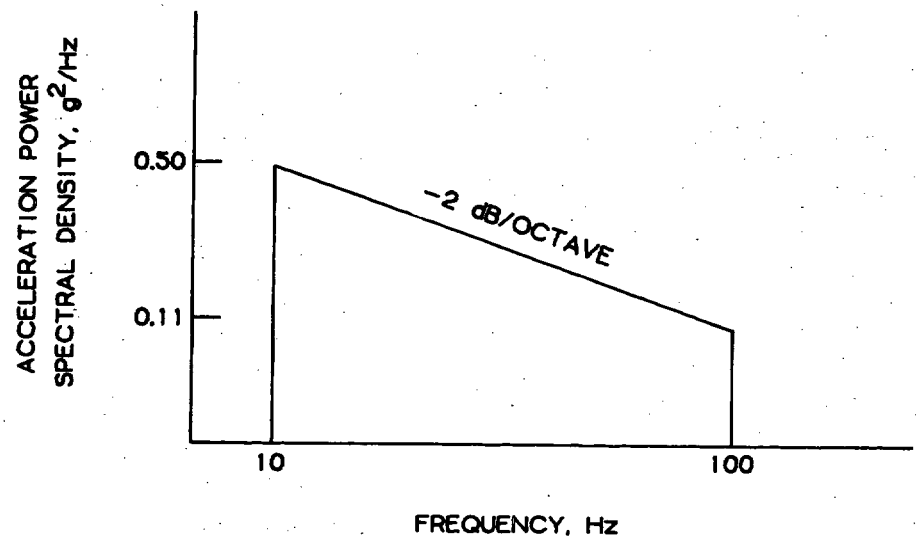


FIGURE 8. Captive flight attack maneuvering (buffeting) random vibration spectrum.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-85744(AS)	FUZE, PROXIMITY, DISPENSER FMU-140/B

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

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